Session 2: Bayesian regression with Big Data

Modern Macroeconometrics

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Day 1 (afternoon session), 7 July 2026

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Overview of Session 2

1. **Kick off this session with some (light) Bayesian basics**: Provide a brief overview of Bayes' Theorem, Bayesian learning and updating, prior and posterior distributions, and predictive inference

2. **Bayesian regression model**: See how general Bayesian theory works in a familiar regression model (connection to classic estimators like OLS)

3. **ML-flavoured shrinkage priors for Big Data**: Stochastic search variable selection (SSVS), Bayesian LASSO (and its frequentist duality), and global-local shrinkage priors

4. **Application**: Which variables predict US GDP growth? A first look at growth at risk, in the spirit of Plagborg-Møller et al. (2020, *Brookings Papers*)

General health warning for the whole course

To keep us on time (and save you from too much technical detail), I skip over most of the Bayesian computational nitty-gritty

Bayesian basics

What does a Bayesian (macro-)econometrician do?

- An econometrician mainly does three things
  1. Estimation of unknown parameters $\boldsymbol{\theta}$ (e.g., regression coefficients) from data $\boldsymbol{y}$
  2. Prediction
  3. Variable selection (e.g., hypothesis testing)

- Bayesian econometrics does these based on a few simple rules of probability in a single coherent framework
  - Begin with general concepts in Bayesian theory before getting to specific models
  - If you know these general concepts you will never get lost

- Bayesian methods for freeing up classical assumptions are relatively straightforward (some of them we will discuss in the next sessions)

Conditional probabilities and Bayes' theorem

- The **conditional probability** of A given B, denoted $\Pr(A \mid B)$, is the probability of event A occurring given that event B has occurred
- For two events A and B, the rules of conditional probability are:

$$\Pr(A \mid B) = \frac{\Pr(A, B)}{\Pr(B)} \qquad \Pr(B \mid A) = \frac{\Pr(A, B)}{\Pr(A)}$$

- Combining these two yields **Bayes' theorem**:

$$\Pr(B \mid A) = \frac{\Pr(A \mid B) \Pr(B)}{\Pr(A)}$$

- This may sound abstract, so let's illustrate this using a data/parameter context

Learning about parameters in a given model (estimation)

- Assume a single model which depends on parameter θ
- Want to figure out properties of the posterior $p(\theta \mid \boldsymbol{y})$
- It is convenient to use Bayes' rule to write the posterior in a different way
- Bayes' rule lies at the heart of Bayesian econometrics:

$$p(B \mid A) = \frac{p(A \mid B)p(B)}{p(A)}$$

- Replace B by θ and A by $\boldsymbol{y}$ to obtain:

$$p(\theta \mid \boldsymbol{y}) = \frac{p(\boldsymbol{y} \mid \theta)p\left(\theta\right)}{p(\boldsymbol{y})}$$

Bayes' theorem: posterior $\propto$ likelihood $\times$ prior

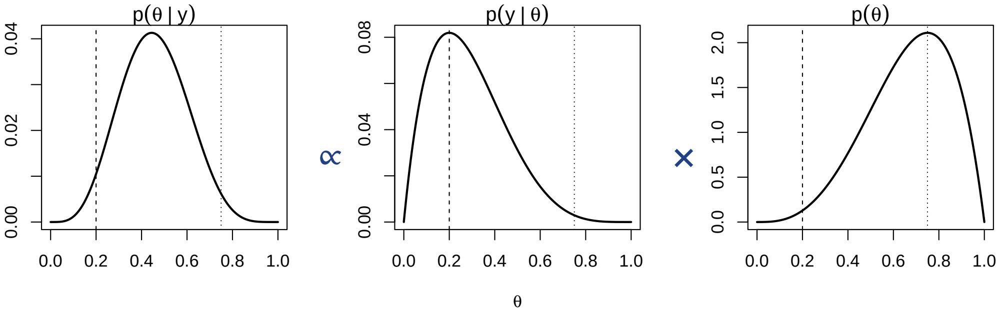
- For estimation can ignore the term $p(\boldsymbol{y})$ since it does not involve θ
- For data $\boldsymbol{y}$ and unknown parameter θ, Bayes' rule gives:

$$\underbrace{p(\theta \mid \boldsymbol{y})}_{\textbf{Posterior}} \propto \underbrace{p(\boldsymbol{y} \mid \theta)}_{\textbf{Likelihood}} \times \underbrace{p(\theta)}_{\textbf{Prior}}$$

- Bayesians treat $p(\theta \mid \boldsymbol{y})$ as being of fundamental interest: "Given the data, what do we know about θ ?"
- Treatment of θ as a random variable is controversial among some econometricians
- **Frequentist econometrics** says that θ is not a random variable

Bayes' theorem in action

We combine data (likelihood) and non-data (prior) information to obtain an updated (proportional) posterior distribution



Prediction in a single model

- Prediction based on the **predictive density** $p(y^* \mid \boldsymbol{y})$
- Since a marginal density can be obtained from a joint density through integration:

$$p(y^* \mid \boldsymbol{y}) = \int p(y^*, \theta \mid \boldsymbol{y}) d\theta$$

- Term inside integral can be rewritten as:

$$p(y^* \mid \boldsymbol{y}) = \int p(y^* \mid \boldsymbol{y}, \theta) p(\theta \mid \boldsymbol{y}) d\theta$$

- Prediction involves the posterior; integrates out parameter uncertainty of θ

Understanding the prior

- The prior $p(\theta)$ is the controversial aspect of Bayesian inference; it sounds unscientific to introduce non-data information
- Bayesian answers:
  1. Often we do have prior information and, if so, we should include it (more information is good)
  2. Can work with so-called "noninformative" priors or a prior that recovers frequentist estimators/properties (see, e.g., LASSO)
  3. Can use hierarchical priors which treat prior hyperparameters as parameters and estimate them; the prior "learns" from the data (see shrinkage priors later on)
  4. Priors obtained from a theoretical model (e.g., a DSGE model)
  5. Prior sensitivity analysis to assess robustness

Bayesian Computation

- Before jumping into the Bayesian regression model, how do we present results from a Bayesian empirical analysis?
- $p(\theta \mid \boldsymbol{y})$ is a pdf; especially if θ is a vector of many parameters cannot present a graph of it
- Want features analogous to frequentist point estimates and confidence intervals
- All these features have integrals in them

Important posterior features

- All of these posterior features have the form

$$\mathbb{E}\left[g\left(\theta\right) \mid \boldsymbol{y}\right] = \int g(\theta) p(\theta \mid \boldsymbol{y}) d\theta,$$

  where $g(\theta)$ is a **function of interest**

- A common point estimate is the mean of the posterior density (or **posterior mean**), obtained with $g\left(\theta\right) = \theta$

- Common measure of dispersion is the **posterior standard deviation** (square root of **posterior variance**), with $\mathbb{V}(\theta \mid \boldsymbol{y}) = \mathbb{E}[\theta^2 \mid \boldsymbol{y}] - \mathbb{E}[\theta \mid \boldsymbol{y}]^2$

- The integrals involved in Bayesian analysis are usually evaluated using **simulation methods**

Intuition of posterior simulation

- Frequentist asymptotic theory uses Laws of Large Numbers (LLN) and Central Limit Theorems (CLT)
- A typical LLN: "consider a random sample, $Y_1, \ldots, Y_N$, as N goes to infinity, the average converges to its expectation" (e.g. $\overline{Y} \longrightarrow \mu$)
- Bayesians use LLN: "consider a random sample from the posterior, $\theta^{(1)}, \ldots \theta^{(S)}$, as S goes to infinity, the average of these converges to $\mathbb{E}[\theta \mid y]$"

Example: Monte Carlo integration

- Let $\theta^{(s)}$ for $s = 1, \ldots, S$ be a random sample from $p(\theta \mid \boldsymbol{y})$ and define

$$\widehat{g}_S = \frac{1}{S} \sum_{s=1}^{S} g\left(\theta^{(s)}\right),$$

  then $\widehat{g}_S$ converges to $\mathbb{E}\left[g(\theta) \mid \boldsymbol{y}\right]$ as S goes to infinity
- **Computational trade-offs**
  - Monte Carlo integration approximates $\mathbb{E}\left[g(\theta) \mid \boldsymbol{y}\right]$, but only if S were infinite would the approximation error be zero
  - We can choose any value for S but larger values of S will increase computational burden
- To gauge size of approximation error, use a CLT to obtain numerical standard error

Bayesian linear regression model

How to approach a regression problem as a Bayesian

- Suppose we want to estimate a simple AR(1) model:

$$y_t = \rho y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2)$$

- We have data $\boldsymbol{y} = (y_1, \ldots, y_T)'$ and two unknown model parameters (ρ and σ^2)
- Assume we know that y_t is a stationary time series (e.g., a growth rate)
- What else do we know about the data? (e.g., how is the data distributed?)
- Given this information, what prior beliefs can we formulate about the AR(1) parameter ρ and the variance parameter σ^2?
- In other words, we wish to specify the prior distributions $p(\rho)$ and $p(\sigma^2)$
  - What do we know about ρ for certain?
  - What do we know about σ^2 for certain?

A regression model

- Let's assume K explanatory variables $x_{t1}, \ldots, x_{tK}$ for $t = 1, \ldots, T$, with $x_{t1} = 1$ for an intercept:

$$y_t = \beta_1 + \beta_2 x_{t2} + \cdots + \beta_K x_{tK} + \varepsilon_t$$

- Stacking over $t = 1, \ldots, T$ gives $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, with

$$\underset{T\times 1}{\boldsymbol{y}} = \begin{bmatrix} y_1 \\ \vdots \\ y_T \end{bmatrix}, \quad \underset{T\times K}{\boldsymbol{X}} = \begin{bmatrix} 1 & x_{12} & \cdots & x_{1K} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{T2} & \cdots & x_{TK} \end{bmatrix}, \quad \underset{K\times 1}{\boldsymbol{\beta}} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_K \end{bmatrix}, \quad \underset{T\times 1}{\boldsymbol{\varepsilon}} = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_T \end{bmatrix}$$

- Classical assumption: $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}_T, \sigma^2 \boldsymbol{I}_T)$

Frequentist vs Bayesian: two views of the same problem

- **Frequentist**: β is a fixed unknown constant (the **truth**); y, and hence $\widehat{\beta}_{\text{OLS}}$, is random, so inference relies on **CLT**:

$$\underbrace{\widehat{\beta}_{\text{OLS}}}_{\text{random}} = (X'X)^{-1}X' \underbrace{y}_{\text{random}} = \underbrace{\beta}_{\text{fixed (true)}} + (X'X)^{-1}X'\varepsilon$$

$$\xrightarrow{\text{CLT}} \widehat{\beta}_{\text{OLS}} \sim \mathcal{N}\left(\beta, \sigma^2 (X'X)^{-1}\right)$$

- **Bayesian**: β is a random variable; combine prior $p(\beta)$ with likelihood $p(y \mid \beta)$ to get posterior $p(\beta \mid y)$; inference via the posterior distribution

- **Key connection**: With a noninformative prior $p(\beta) \propto \text{const}$, the Bayesian posterior mean = $\widehat{\beta}_{\text{OLS}}$

Parameters of interest and the Big Data challenge

- Parameters of interest: $K \times 1$ coefficient vector $\boldsymbol{\beta}$ and the error variance σ^2
- **OLS**: $\widehat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$
- In the **Big Data** context K can be large; comparable to or even larger than T
- OLS **fails** when $K \geq T$: $\boldsymbol{X}'\boldsymbol{X}$ is not invertible
- **Bayesian view**: Incorporate prior and shrinkage information when K is large

The likelihood function

- Likelihood can be derived under the classical assumptions
- All elements of X are fixed (i.e., not random variables)
- OLS quantities given by:

$$\widehat{\beta} = (X'X)^{-1} X'y, \quad \nu = T - K, \quad s^2 = \frac{\left(y - X\widehat{\beta}\right)' \left(y - X\widehat{\beta}\right)}{\nu}$$

- Likelihood function can be written in terms of OLS quantities

$$p(y \mid \beta, \sigma^2) = \frac{1}{(2\pi)^{\frac{T}{2}}} \underbrace{\exp\left[-\frac{1}{2\sigma^2}\left(\beta - \widehat{\beta}\right)' X'X \left(\beta - \widehat{\beta}\right)\right]}_{\text{multivariate Normal in } \beta} \times \underbrace{\left(\sigma^2\right)^{-\frac{T}{2}} \exp\left[-\frac{1}{2\sigma^2} \nu s^2\right]}_{\text{inverse Gamma in } \sigma^2}$$

Choosing a prior

Natural conjugate prior

- Common starting point is natural conjugate prior
- β conditional on σ^2 is **multivariate Normal** and prior for error variance σ^2 is **inverse Gamma**:

$$\beta \mid \sigma^2 \sim \mathcal{N}(\underline{\beta}, \sigma^2 \underline{V})$$
$$\sigma^2 \sim \mathcal{IG}(\underline{a}, \underline{b})$$

- **Hyperparameters** chosen by the researcher:
  - $\underline{\beta}$: Prior mean for the coefficients
  - $\underline{V}$: Prior covariance matrix (scales with σ^2)
  - $\underline{a}$: Prior shape parameter for the error variance
  - $\underline{b}$: Prior scale parameter for the error variance

The joint posterior of β and σ^2

- Multiplying likelihood by prior and collecting terms, the posterior can be shown to have the same form as the prior
- The conditional posterior for β (conditional on σ^2) is **multivariate Normal** and the posterior for error variance σ^2 is **inverse Gamma**:

$$\beta \mid y, \sigma^2 \sim \mathcal{N}\left(\overline{\beta}, \sigma^2 \overline{V}\right)$$

with

$$\overline{V} = \left(\underline{V}^{-1} + X'X\right)^{-1}$$

$$\overline{\beta} = \overline{V}\left(\underline{V}^{-1}\underline{\beta} + X'X\widehat{\beta}\right)$$

$$\sigma^2 \mid y \sim \mathcal{IG}(\overline{a}, \overline{b})$$

with

$$\overline{a} = \underline{a} + \tfrac{T}{2}$$

$$\overline{b} = \underline{b} + \tfrac{1}{2}\left(y'y + \underline{\beta}'\underline{V}^{-1}\underline{\beta} - \overline{\beta}'\overline{V}^{-1}\overline{\beta}\right)$$

The marginal posterior of $\boldsymbol{\beta}$ and predictive density

- The marginal posterior for $\boldsymbol{\beta}$ is obtained by integrating out σ^2
- It turns out that $\boldsymbol{\beta}$ is multivariate t-distributed:

$$\boldsymbol{\beta} \mid \boldsymbol{y} \sim t\left(\overline{\boldsymbol{\beta}}, \frac{\overline{b}\,\overline{\boldsymbol{V}}}{\overline{a}}, 2\overline{a}\right)$$

- **Predictive density** for new observation y^* given regressors $\boldsymbol{x}^*$:

$$p(y^* \mid \boldsymbol{x}^*, \boldsymbol{y}) = \int p(y^* \mid \boldsymbol{x}^*, \boldsymbol{\beta}, \sigma^2)\, p(\boldsymbol{\beta}, \sigma^2 \mid \boldsymbol{y})\, d\boldsymbol{\beta}\, d\sigma^2$$

- It turns out the predictive density is also a t distribution, automatically capturing parameter uncertainty in the forecast

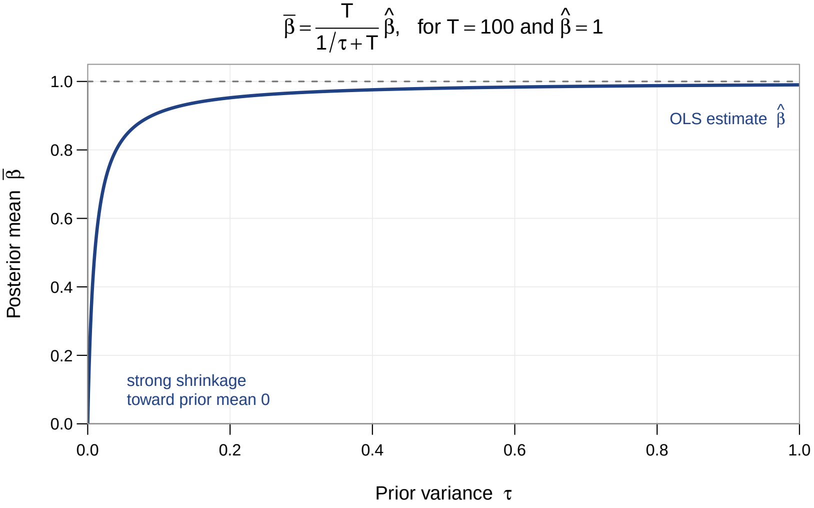
What does a prior do?

- To understand the idea of shrinkage, let's illustrate what the prior does
  - $\sigma^2 = 1$ is known and β is a scalar
  - $\boldsymbol{X} = (1, \ldots, 1)'$, such that $\boldsymbol{X}'\boldsymbol{X} = T$
  - $\underline{V} = \tau$ and $\underline{\beta} = 0$
- In this case the posterior is:

$$\overline{\beta} = \frac{T}{\frac{1}{\tau} + T}\widehat{\beta}$$

- Think of the extreme cases ($\tau \longrightarrow \infty$ and $\tau \longrightarrow 0$)
  - If $\tau \longrightarrow \infty$, then $\frac{1}{\tau} \longrightarrow 0$ and $\frac{T}{\frac{1}{\tau}+T} \longrightarrow 1$, so $\overline{\beta} \longrightarrow \widehat{\beta}$ (i.e., **OLS**)
  - If $\tau \longrightarrow 0$, then $\frac{1}{\tau} \longrightarrow \infty$ and $\frac{T}{\frac{1}{\tau}+T} \longrightarrow 0$, so $\overline{\beta} \longrightarrow 0$ (the prior mean, i.e., **exclusion**)

Implied prior shrinkage by varying τ



Prior shrinkage

- Posterior mean is pulled towards zero (“shrinkage")
- Commonly done to avoid over-fitting/over-parameterisation problems
- Strength of prior shrinkage controlled through prior variance
  - If τ is small, then strong prior information β is near 0
  - If τ is big, then prior becomes more noninformative see Appendix
- Note: exactly what “small” and “large” means depends on the empirical application and units of measurement of data

Shrinkage priors for Big Data

Some context: Big Data regression problem

- Re-consider a regression with K potential predictors: $y_t = \boldsymbol{x}_t'\boldsymbol{\beta} + \varepsilon_t$
- Most predictors are likely irrelevant, but a priori we do not know which ones (and the degree of relevance varies across them; see Session 1)
- A noninformative prior cannot discriminate between relevant and irrelevant variables
- We need priors that:
  - **Shrink some irrelevant predictors towards zero** to avoid overfitting
  - **Keep some relevant predictors close to OLS** to preserve signal

Some motivation: Growth at risk for later on

- **Growth at risk**: Forecast the **conditional distribution** of future GDP growth, especially its **lower tail** (downside risk; see Session 5), from many macro-financial predictors (e.g., FRED-QD)
- **Why this is hard**: Many predictors K, short macro samples T $(K > T)$; the **tail** is identified by only a few left-tail observations
- But before we turn to tail behaviour and quantiles, we first work through the standard linear regression (the conditional mean)
- Plagborg-Møller et al. (2020, *Brookings Papers*) reassess the GaR story and find that financial variables have very limited predictive power; signals are driven by a handful of episodes

Interpretation of Bayesian shrinkage

- The posterior mean $\overline{\beta}$ is a **weighted average** of the prior mean $\underline{\beta}$ and the OLS estimator $\widehat{\beta}$:

$$\overline{\beta} = \underbrace{\overline{V}\underline{V}^{-1}}_{\text{prior weight}} \underline{\beta} + \underbrace{\overline{V}X'X}_{\text{data weight}} \widehat{\beta}$$

- Setting $\underline{\beta} = \mathbf{0}$ implies the posterior mean is **shrunk towards zero**
  - Small prior variance $\underline{V}$: strong shrinkage, $\overline{\beta} \approx \mathbf{0}$
  - Large prior variance $\underline{V}$: weak shrinkage, $\overline{\beta} \approx \widehat{\beta}$

Non-conjugate priors

- Normal linear regression with **natural conjugate prior** $\Rightarrow$ posterior and predictive distributions available in **closed form**
- Shrinkage priors / more complex models $\Rightarrow$ **no analytical results**
- Extensions to shrinkage priors of the regression model requiring a **Gibbs sampler** for posterior computation see Appendix
  - **Gibbs sampler explained in a sentence**: Draw each parameter block ($\boldsymbol{\beta}$, then σ^2) from its **full conditional posterior** in turn; cycling through these draws approximates the joint posterior
  - Keep the Normal linear regression model (under the classical assumptions) as before
  - Likelihood function same as before
  - Parameters of model will still be $\boldsymbol{\beta}$ and σ^2
  - We then only introduce flexibility through the prior

The independent Normal-Inverse-Gamma prior

- Before we had conjugate prior where $p\left(\boldsymbol{\beta} \mid \sigma^2\right)$ was Normal density and $p\left(\sigma^2\right)$ inverse Gamma density
- Now use similar prior, but assume prior **independence** between $\boldsymbol{\beta}$ and σ^2

$$p\left(\boldsymbol{\beta}, \sigma^2\right) = p\left(\boldsymbol{\beta}\right) p\left(\sigma^2\right)$$

- $p\left(\boldsymbol{\beta}\right)$ is Normal and $p\left(\sigma^2\right)$ inverse Gamma:

$$\boldsymbol{\beta} \sim \mathcal{N}\left(\underline{\boldsymbol{\beta}}, \underline{\boldsymbol{V}}\right)$$
$$\sigma^2 \sim \mathcal{IG}(\underline{a}, \underline{b})$$

- Key difference: $\underline{\boldsymbol{V}}$ is now the prior covariance matrix of $\boldsymbol{\beta}$ directly; with the conjugate prior we had $\mathbb{V}(\boldsymbol{\beta} \mid \sigma^2) = \sigma^2 \underline{\boldsymbol{V}}$

The posterior

- Posterior $\propto$ prior $\times$ likelihood; **joint posterior** for (β, σ^2) has **no closed form**
- But the two **conditional posteriors** are simple:

$$\beta \mid y, \sigma^2 \sim \mathcal{N}\left(\overline{\beta}, \overline{V}\right)$$

with

$$\overline{V} = \left(\underline{V}^{-1} + \sigma^{-2} X'X\right)^{-1}$$

$$\overline{\beta} = \overline{V}\left(\underline{V}^{-1}\underline{\beta} + \sigma^{-2} X'y\right)$$

$$\sigma^2 \mid y, \beta \sim \mathcal{IG}\left(\overline{a}, \overline{b}\right)$$

with

$$\overline{a} = \underline{a} + \tfrac{T}{2}$$

$$\overline{b} = \underline{b} + \tfrac{1}{2}(y - X\beta)'(y - X\beta)$$

- **Key point**: the conditionals do not directly characterize the joint posterior $\Rightarrow$ use a **Gibbs sampler** see Appendix

Popular shrinkage priors

Stochastic Search Variable Selection (SSVS)

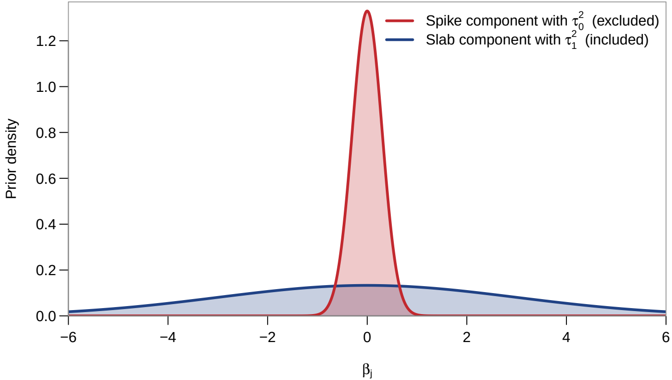
George & McCulloch (1993, *JASA*)

- A **spike and slab prior** on each coefficient β_j:

$$\beta_j \mid \gamma_j \sim (1-\gamma_j)\,\mathcal{N}(0,\tau_0^2) + \gamma_j\,\mathcal{N}(0,\tau_1^2)$$

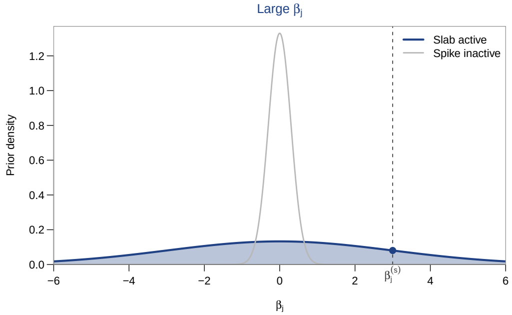
- Binary indicator $\gamma_j \in \{0,1\}$ controls whether x_j is **included or excluded**:
  - $\gamma_j = 0$: **Spike** active (τ_0^2 very small) $\Rightarrow \beta_j \approx 0$ (variable x_j excluded)
  - $\gamma_j = 1$: **Slab** active (τ_1^2 large) $\Rightarrow \beta_j$ unconstrained (variable x_j included)
- Prior on inclusion: $\gamma_j \sim \text{Bernoulli}(\pi)$ with π typically set to $1/2$ (equal prior probability of inclusion or exclusion)
- The Gibbs sampler draws $\boldsymbol{\beta}$, σ^2, and $\boldsymbol{\gamma}$ iteratively from well-known conditional posteriors

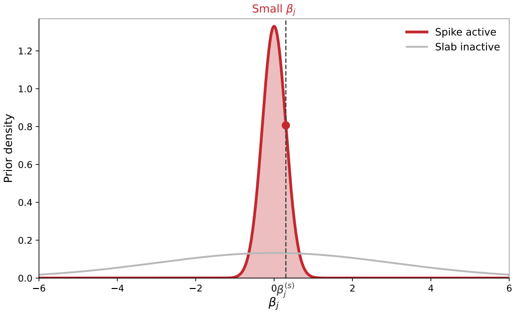
SSVS: The spike and slab components illustrated



- **Spike** ($\gamma_j = 0$): Mass concentrated at 0 (excluded)
- **Slab** ($\gamma_j = 1$): Diffuse/flat (included)

SSVS: When is each component active?





- At draw s: A **large** $\beta_j^{(s)}$ is explained only by the **slab** ($\gamma_j = 1$, included); a **small** $\beta_j^{(s)}$ by the **spike** ($\gamma_j = 0$, excluded)
- The Gibbs sampler updates γ_j via the inclusion probability $p(\gamma_j^{(s)} = 1 \mid \beta_j^{(s)})$

SSVS: Interpretation of posterior inclusion probability

- Posterior $p(\gamma_j = 1 \mid y)$: **posterior inclusion probability (PIP)** tells us whether x_j matters
- Provides a natural **variable selection summary** and effectively allows for hypothesis testing
  - PIP > 0.5: variable j is likely relevant (**median probability model**)
  - PIP < 0.05: strong evidence that variable j can be excluded
- Amounts to **Bayesian model averaging (BMA)** with the posterior integrating over all 2^K possible models (accounts for model uncertainty)
- A **gold standard** for variable selection, but it breaks down in large settings ($K > 40$)

LASSO: Bayesian and Frequentist view

- The classic **LASSO** (Tibshirani, 1996, *JRSS:B*) minimises

$$\widehat{\beta}_{\text{LASSO}} = \arg\min_{\boldsymbol{\beta}} \left\{ \sum_{t=1}^{T} (y_t - \boldsymbol{x}_t' \boldsymbol{\beta})^2 + \lambda \sum_{j=1}^{K} |\beta_j| \right\}$$

- **Bayesian LASSO** (Park & Casella, 2008, *JASA*): Penalty can be represented as **Laplace prior** $p(\beta_j) \propto \exp(-\lambda |\beta_j| / \sigma)$
- The classic LASSO estimate is the **posterior mode** under this prior
- **Bayesian** integrates over the full posterior, not just point estimate

LASSO: A hierarchical prior view

- **Scale mixture representation** allows Gibbs sampling

$$\beta_j \mid \tau_j^2 \sim \mathcal{N}(0, \sigma^2 \tau_j^2), \qquad \tau_j^2 \sim \text{Exp}(\lambda^2/2)$$

  with each τ_j^2 being a **local** (predictor-specific) prior variance

- As we should know by now, a small τ_j^2 shrinks β_j towards zero
- **But**: LASSO tends to over-shrink signals
- Motivates more sophisticated, so-called **global-local shrinkage priors**
- Key idea is nicely summarised in the title of the paper *"Shrink Globally, Act Locally,"* Polson and Scott (2011)

Global-local shrinkage priors

- Modern shrinkage priors use **global** and **local** scales:

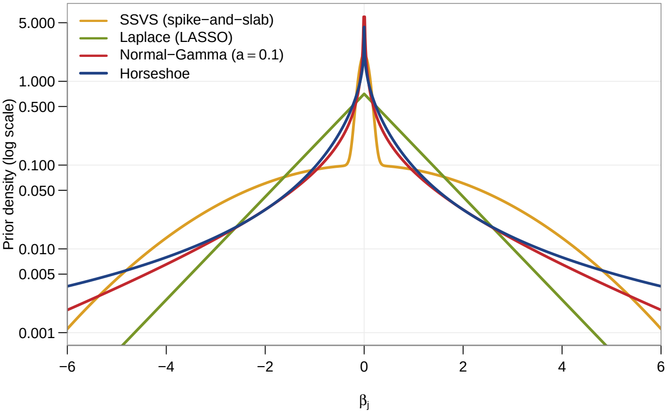
$$\beta_j \mid \phi_j^2, \lambda^2 \sim \mathcal{N}(0, \phi_j^2 \lambda^2)$$

- $\lambda > 0$: **Global shrinkage parameter** controls overall level of shrinkage
- $\phi_j > 0$: **Local shrinkage parameter** allows individual coefficients to escape the global shrinkage
- A good prior should **heavily shrink most coefficients** but **not overshrink the few large ones**
- **Shrinkage factor** $\kappa_j = \frac{1}{1+\phi_j^2\lambda^2} \in (0, 1)$
  - **Noise** ($|\beta_j| \approx 0$): $\kappa_j \to 1$, full shrinkage to zero
  - **Signal** ($|\beta_j|$ large): $\kappa_j \to 0$, (nearly) unshrunk

Global-local shrinkage priors (cont.)

- Different choices of prior on ϕ_j and λ define different shrinkage priors:
  - Horseshoe: $\phi_j \sim C^+(0,1)$, $\lambda \sim C^+(0,1)$
  - Normal-Gamma: $\phi_j^2 \sim \mathcal{G}(\vartheta, \lambda^2/2)$, $\lambda^2 \sim \mathcal{G}(a_\lambda, b_\lambda)$
  - LASSO: special case of Normal-Gamma with $\vartheta = 1$
- **Efficient Gibbs sampler** available for all these variants see Appendix
- **Current gold standard** for continuous shrinkage in high-dimensional regression; priors adapt automatically to both sparse and dense signals
- Minimal input of the researcher required (tuning-free) and hence basically ML!
- Downside: No PIPs, but approximation possible with *"first shrink, then sparsify!"* (see, e.g., Huber, Koop, and Onorante, 2021 *JBES*)

Shrinkage profiles of the different priors



A good shrinkage prior has a **spike at** 0 (shrinks noise) and **heavy tails** (keeps large signals)

Application

When is growth at risk?

Application: When is growth at risk?

Plagborg-Møller et al. (2020, *Brookings Papers*)

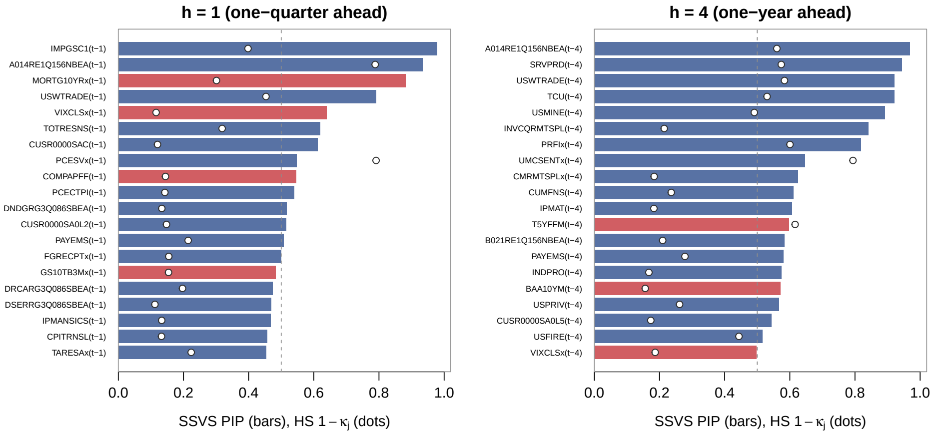
- Do financial variables predict (the **distribution** of) GDP growth?
- **Data**: FRED-QD augmented with key financial indicators (NFCI components); multi-country extension covering **13 advanced economies**
- Variable selection and shrinkage priors are **key**
- **Main finding**: Moments beyond the conditional mean are **poorly estimated**
  - Real indicators dominate (inventories, income, employment, housing, imports)
  - Only two financial variables enter robustly: implied volatility (`VXO`) and the AAA-to-Treasury spread (`AAASPR`)
- Financial variables carry **little predictive power** for tail risks

Setting up a similar variable selection exercise

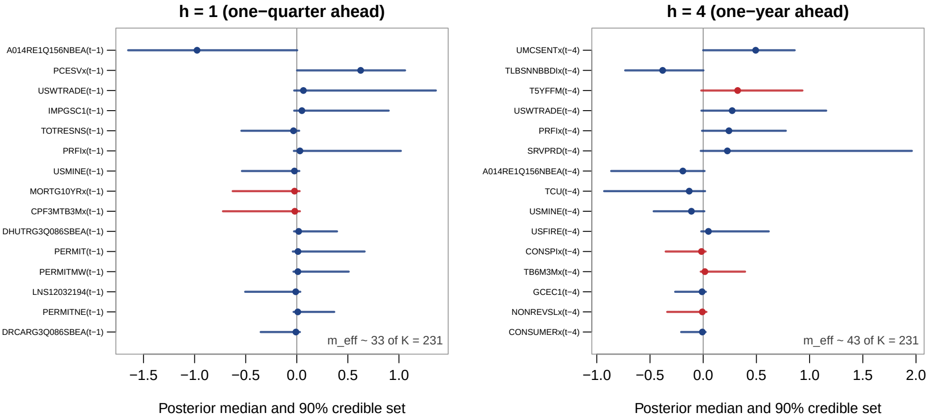
Plagborg-Møller et al. (2020, *Brookings Papers*)

- Revisit the paper's US variable selection exercise with this session's tools
- Focus on FRED-QD and conditional mean
- **Data**: FRED-QD (2026-04 vintage); balanced panel of $K = 231$ standardised predictors (plus an intercept), 1973Q1–2019Q4 ($T = 187$ for $h = 1$) $\Rightarrow$ OLS infeasible
- **Regression setup**
  - Target: Annualised GDP growth, $y_t = \frac{400}{h}\left(\log \mathrm{GDP}_t - \log \mathrm{GDP}_{t-h}\right)$ for $h \in \{1, 4\}$
  - K predictors $\boldsymbol{x}_{t-h}$ (including intercept)
- **Priors**: SSVS (gives PIPs) and horseshoe (gives shrinkage factors κ_j), with the Gibbs samplers sketched in the appendix

Application: Which predictors are selected?



Application: Few signals escape the shrinkage



Appendix

A noninformative prior

- Noninformative prior sets $\underline{a} = 0$ and $\underline{V}$ is big (implies large prior uncertainty)
- A common way: $\underline{V}^{-1} = \tau^{-1} \boldsymbol{I}_K$ where τ^{-1} is a scalar and let τ^{-1} go to zero
- This noninformative prior is improper and becomes:

$$p\left(\boldsymbol{\beta}, \sigma^2\right) \propto \frac{1}{\sigma^2}$$

- With this choice we get OLS results:

$$\boldsymbol{\beta} \mid \sigma^2, \boldsymbol{y} \sim \mathcal{N}\left(\overline{\boldsymbol{\beta}}, \sigma^2 \overline{\boldsymbol{V}}\right)$$

with $\overline{\boldsymbol{V}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}$ and $\overline{\boldsymbol{\beta}} = \widehat{\boldsymbol{\beta}}$

$$\sigma^2 \mid \boldsymbol{y} \sim \mathcal{IG}(\overline{a}, \overline{b})$$

with $\overline{a} = \frac{T}{2}$ and $\overline{b} = \frac{1}{2}\left(\nu s^2\right)$

Popular MCMC algorithms

The Gibbs sampler

- The **Gibbs sampler** uses conditional posteriors to draw $\boldsymbol{\beta}^{(s)}$ and $(\sigma^2)^{(s)}$ for $s = 1, \ldots, S$
- Averages yield posterior estimates, as with Monte Carlo integration
- Powerful and widely-used tool for posterior simulation in econometric models
- **General setup:** $\boldsymbol{\theta}$ is a p-vector of parameters with likelihood $p(\boldsymbol{y} \mid \boldsymbol{\theta})$, prior $p(\boldsymbol{\theta})$ and posterior $p(\boldsymbol{\theta} \mid \boldsymbol{y})$
- Partition $\boldsymbol{\theta}$ into B **blocks**:

$$\boldsymbol{\theta} = \left(\boldsymbol{\theta}'_{(1)}, \boldsymbol{\theta}'_{(2)}, \ldots, \boldsymbol{\theta}'_{(B)}\right)'$$

- For example, regression model: $B = 2$ with $\boldsymbol{\theta}_{(1)} = \boldsymbol{\beta}$ and $\boldsymbol{\theta}_{(2)} = \sigma^2$

Intuition of the Gibbs sampler

- **Monte Carlo integration**: Draw from $p(\boldsymbol{\theta} \mid \boldsymbol{y})$ and average to estimate $\mathbb{E}[g(\boldsymbol{\theta}) \mid \boldsymbol{y}]$ for any function $g(\boldsymbol{\theta})$
- **Problem:** Drawing directly from $p(\boldsymbol{\theta} \mid \boldsymbol{y})$ is often **not possible** (e.g., not available in closed form)
- But it often is easy to draw from **full conditional posteriors**:

$$p(\boldsymbol{\theta}_{(b)} \mid \boldsymbol{y}, \{\boldsymbol{\theta}_{(j)}\}_{j \neq b}), \quad b = 1, \ldots, B$$

- **Gibbs sampler**: Cycle through draws from each full conditional; in turn, produces a sequence $\boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)}, \ldots, \boldsymbol{\theta}^{(S)}$ that can be averaged to estimate $\mathbb{E}[g(\boldsymbol{\theta}) \mid \boldsymbol{y}]$, just as with Monte Carlo integration

Gibbs sampler for Normal-Gamma

- The Normal-Gamma mixture representation enables **data augmentation** in a Gibbs sampler with well known conditionals
- For a regression with $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \boldsymbol{I}_T)$, iterate through:

1. Draw $\boldsymbol{\beta} \mid \bullet \sim \mathcal{N}\left(\overline{\boldsymbol{\beta}}, \overline{\boldsymbol{V}}\right)$, with $\overline{\boldsymbol{V}} = \left(\underline{\boldsymbol{V}}^{-1} + \sigma^{-2}\boldsymbol{X}'\boldsymbol{X}\right)^{-1}$, $\overline{\boldsymbol{\beta}} = \sigma^{-2}\overline{\boldsymbol{V}}\boldsymbol{X}'\boldsymbol{y}$, and $\underline{\boldsymbol{V}} = \lambda^2 \operatorname{diag}(\phi_1^2, \ldots, \phi_K^2)$
2. Draw $\sigma^2 \mid \bullet \sim \mathcal{IG}\left(\overline{a}, \overline{b}\right)$, with $\overline{a} = \underline{a} + \frac{T}{2}$ and $\overline{b} = \underline{b} + \frac{1}{2}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$
3. Draw $\phi_j^2 \mid \bullet \sim \mathcal{GIG}\left(\vartheta - \frac{1}{2},\ \lambda^2,\ \beta_j^2\right)$ for each $j = 1, \ldots, K$
4. Draw $\lambda^2 \mid \bullet \sim \mathcal{G}\left(a_\lambda + K\vartheta,\ b_\lambda + \frac{1}{2}\sum_{j=1}^{K} \phi_j^2\right)$

Gibbs sampler for the Horseshoe

- The **inverse Gamma augmentation** of Makalic & Schmidt, 2016, *IEEE* (auxiliaries ν_j, ξ) results in a Gibbs sampler with well known conditionals
- For $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, iterate through:
  1. Draw $\boldsymbol{\beta} \mid \bullet \sim \mathcal{N}\left(\overline{\boldsymbol{\beta}}, \overline{\boldsymbol{V}}\right)$, with $\overline{\boldsymbol{V}} = \left(\underline{\boldsymbol{V}}^{-1} + \sigma^{-2}\boldsymbol{X}'\boldsymbol{X}\right)^{-1}$, $\overline{\boldsymbol{\beta}} = \sigma^{-2}\overline{\boldsymbol{V}}\boldsymbol{X}'\boldsymbol{y}$, and $\underline{\boldsymbol{V}} = \lambda^2 \operatorname{diag}(\phi_1^2, \ldots, \phi_K^2)$
  2. Draw $\sigma^2 \mid \bullet \sim \mathcal{IG}\left(\overline{a}, \overline{b}\right)$, with $\overline{a} = \underline{a} + \frac{T}{2}$ and $\overline{b} = \underline{b} + \frac{1}{2}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$
  3. Draw $\phi_j^2 \mid \bullet \sim \mathcal{IG}\left(1,\ \frac{1}{\nu_j} + \frac{\beta_j^2}{2\lambda^2}\right)$ and $\nu_j \mid \phi_j^2 \sim \mathcal{IG}\left(1,\ 1 + \frac{1}{\phi_j^2}\right)$ for each j
  4. Draw $\lambda^2 \mid \bullet \sim \mathcal{IG}\left(\frac{K+1}{2},\ \frac{1}{\xi} + \sum_{j=1}^K \frac{\beta_j^2}{2\phi_j^2}\right)$ and $\xi \mid \lambda^2 \sim \mathcal{IG}\left(1,\ 1 + \frac{1}{\lambda^2}\right)$